

M.C.A. (Two Years Course) First Year First Semester (CBCS)
1T2 Paper-II - Data Communication and Network

P. Pages : 2

Time : Three Hours



KRS/KS/25/1779

Max. Marks : 80

- Notes : 1. All questions are compulsory and carry equal marks.
2. Draw well labelled diagram wherever necessary.

Either :

1. a) Explain the OSI reference model with a diagram. 8
b) Explain transmission and switching in physical layer. 8

OR

- c) What is ISDN? Explain the interfaces supported by ISDN. 8
d) Write note on: 8
i) Protocol specification and verification
ii) Protocols performance.

Either :

2. a) Explain SMTP and POP3 protocols in details. 8
b) What is cryptography? Discuss its techniques and types with any two examples. 8

OR

- c) What is RPC? How the Remote Procedure Call (RPC) executes in the session Layer? Explain. 8
d) Explain the connection management of TCP. 8

Either :

3. a) Describe triple DES and its applications. 8
b) Write note on: 8
i) Passive attacks
ii) Active attacks.

OR

- c) Explain Network security model and its vulnerabilities. 8
d) Explain IDEA algorithm in detail. 8

Either :

4. a) Write note on: 8
- i) Packet filtering.
- ii) Access control.
- b) Discuss the RSA Diffie – Hellman in detail. 8

OR

- c) What is Authentication? Explain Password based authentication. 8
- d) What is firewalls? Why firewall used? Explain in detail. 8
5. a) Explain Sliding window protocols. 4
- b) Explain UDP protocol in detail. 4
- c) Write advantages and disadvantages of Intruders. 4
- d) Explain the Authentication certificate in detail. 4
